

Secure Operations of Connected and Autonomous Vehicles

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Abstract—With features of collaborative interaction and autonomous decision-making, connected and autonomous vehicles (CAVs) offer a viable solution for a sustainable and efficient future of transportation. However, with the development of CAVs, questions have been raised about weaknesses in cybersecurity and autonomous driving capabilities. In particular, CAVs provide new opportunities for cyberattackers, posing a significant threat to future road and vehicle data security. Therefore, this paper provides a comprehensive analysis of the cybersecurity environment of CAVs. First, we propose a systematic classification of six security threats from cybersecurity principles: false data, information theft, privilege escalation, block communication, and time delay. We review the cybersecurity environment of CAVs from a lifecycle perspective, covering aspects from development to accident, and present the existing cybersecurity countermeasures. These involve security standards designation, verification and validation of vehicle networks, resilient strategies for self-defined driving, and attack detection and digital forensics. Finally, based on our systematic review, we propose a conceptual vehicle security operations center (VSOC) framework that provides valuable inspiration and references for future research and industrial applications for CAVs cybersecurity.

Index Terms—Connected and autonomous vehicles, cybersecurity, vehicle security operation center.

I. INTRODUCTION

AUTONOMOUS vehicles (AVs) are classified into six levels based on their autonomous driving capabilities [1], [2]. Vehicle-to-everything communication (V2X) is necessary for AVs to achieve higher levels of autonomous driving capabilities. V2X brings collaborative sensing, control, and decision-making to

AVs, giving rise to the concept of connected and autonomous vehicles (CAVs). The adoption of CAVs leads to many safe and sustainable benefits. It reduces human-caused traffic accidents, vehicle congestion time, and carbon emissions [3], [4]. In addition, the labor force released from vehicle driving will support the labor market in the aging trend and augment social wealth. However, CAVs are still in the exploration stage of new technologies and lack cognition of potential new threats [5]. Only when the technology is ready will the actual threat range be revealed.

Open-access communication systems provide easy collaboration for CAVs and make them vulnerable to various cyberattacks. These cyberattacks involve collecting the information about the vehicle and its user and they can also potentially gain control over the vehicle itself [6]. The methods include spoofing attacks, replay attacks, man-in-the-middle attacks, and denial-of-service attacks. Specifically, the ability of CAVs to drive autonomously relies on accurate GPS signals; when an attacker misleads the vehicle to trust the wrong location to generate the desired driving route, causing it to arrive at the mistaken destination [7]. Moreover, attack methods such as malware in the infotainment system and malicious nodes in the communication network threaten user information security. As the technology of CAVs increases, cybersecurity countermeasures are evolving with continuous improvements. Generic and proven solutions are released as standards to implement it in the industry. The encryption and verification technology of vehicle communication verifies the authenticity of transmitted data. Table I provides a summary of the research work, facilitating a closer look at cybersecurity threats and countermeasures [8], [9], [10], [11], [12], [13], [14], [15], [16].

This work aims to present a systematic top-down classification system for cybersecurity threats and countermeasures. Our contributions to the field are summarized in the following three key aspects:

- This work focuses on expanding five categories of cybersecurity threats: block communication (BC), time delay (TD), false data (FD), information theft (IT), and privilege escalation (PE). These categories are derived from the security properties of confidentiality, integrity, and availability (CIA). Additionally, the article presents a summary of various cyberattack methods for each of these threat categories;
- Our analysis of cybersecurity countermeasures for CAVs focuses on their lifecycle, encompassing three distinct

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TABLE I
SUMMARY OF RELATED WORK

Paper		[8]	[9]	[10]	[11]	[12]	[13]	[14]	[15]	[16]	This Paper
Threats	False Data	✓	✓	✓	✓	✓	✓	✓	✓		✓
	Information Theft		✓		✓		✓		✓		✓
	Privilege Escalation								✓		✓
	Block Communication	✓	✓	✓	✓	✓	✓	✓	✓		✓
	Time Delay			✓		✓		✓			✓
Countermeasures	Standardization Efforts	✓			✓			✓			✓
	Authentication and Verification	✓	✓		✓		✓	✓			✓
	Resilient Strategies					✓					✓
	Attack Detection	✓	✓			✓	✓				✓
	Digital Forensics									✓	✓
Operation Framework	Vehicle Security Operations Center										✓

phases: development, operation, and accidental end-of-life. In the development phase, defining important rules and protocols like standard effort (SE), Authentication and Verification (A&V) crucial. During the operation phase, resilient strategy (RS) and attack detection (AD) emerge as real-time protection methods. Furthermore, digital forensics (DF) plays a critical role in an effective investigation feedback mechanism following an accident;

- We propose a parallel security-based architecture for vehicle security operations, including four key components: management, orchestration, detection, and response (MODR).

The remainder of this article is structured as follows: Section II introduces the CAV and vehicle operation center (VOC), which are essential components of the CAVs system framework. The potential cybersecurity threats and taxonomy for CAVs are provided in Section III. Section IV describes the cybersecurity countermeasures for existing CAVs. Section V presents the composition and mechanism of the VSOC framework, while Section VI provides the conclusion of this work.

II. OVERVIEW OF CONNECTED AND AUTONOMOUS VEHICLES

For the impact of cybersecurity, previous studies focused on the vehicle or platoon, and the perspective of this article is the vehicle environment constituted by CAVs. As depicted in Fig. 1, the traffic environment of CAVs mainly consists of CAVs, VOC and VSOC, which will be introduced as follows, mainly CAVs and VOC, and VSOC in Section V.

A. Connected and Autonomous Vehicle

CAV combines vehicle connectivity and automation, and its prospect of enhancing traffic safety and efficiency has sparked the attention of research and industry. Researchers are not only exploring the highly autonomous driving environment of purely CAV but also discussing the mixed traffic environment where CAVs and human-driven vehicles (HDVs) coexist [17], [18]. This article focuses on the former's secure operational threats

and countermeasures. In the highly autonomous driving environment, traffic scenarios such as vehicle queues, ramps, intersections, and lane changes are the application areas of autonomous driving functions. For CAV, vehicle perception, communication, and control are essential to complete autonomous driving abilities.

1) *Perception*: While traditional autonomous vehicles (AVs) derive their perception from on-board sensors, V2X and VOC in CAVs will not only help vehicles obtain information around the vehicle but also provide information that is difficult to obtain. On-board sensors provide the basic sensing and cognitive capabilities for autonomous vehicles. The primary on-board sensors include ultrasonic radar, mm-wave radar, LiDAR [19], camera and GPS. Simultaneously, the comprehensive capability of AV for environmental perception is enhanced by overcoming the defects of a single sensor through sensing data fusion technology. The development of connected vehicles has expanded the boundaries of AVs' sensing capabilities. CAVs realize the interaction of sensory information between vehicles and vehicles (V2V), infrastructure (V2I), pedestrians (V2P), and other traffic participants through V2X and other communication technologies [20], [21], enabling CAVs to have more comprehensive and abundant information about the driving environment. Furthermore, VOC provides operational support for CAVs, serves in the lifecycle of CAVs, and balances social supply and demand for CAVs.

2) *Communication*: The communication network of CAVs can be divided into intra-vehicle and inter-vehicle networks, a comprehensive system network for communication and information exchange in and out of the vehicle per the agreed communication protocol and data interaction standard. Each CAV has its intra-vehicle communication network, such as the controller area network (CAN), local interconnect network (LIN), media oriented system transport (MOST) and FlexRay, to enable communication between in-vehicle electronic control units (ECUs). The inter-vehicle network is made up of ubiquitous communication terminals such as V2V, V2I, and V2P, also known as V2X [22]. V2X evolved from the early dedicated short-range communication (DSRC) to cellular V2X (C-V2X), and satellite communication methods in some special intelligent applications gave birth to satellite V2X (S-V2X).

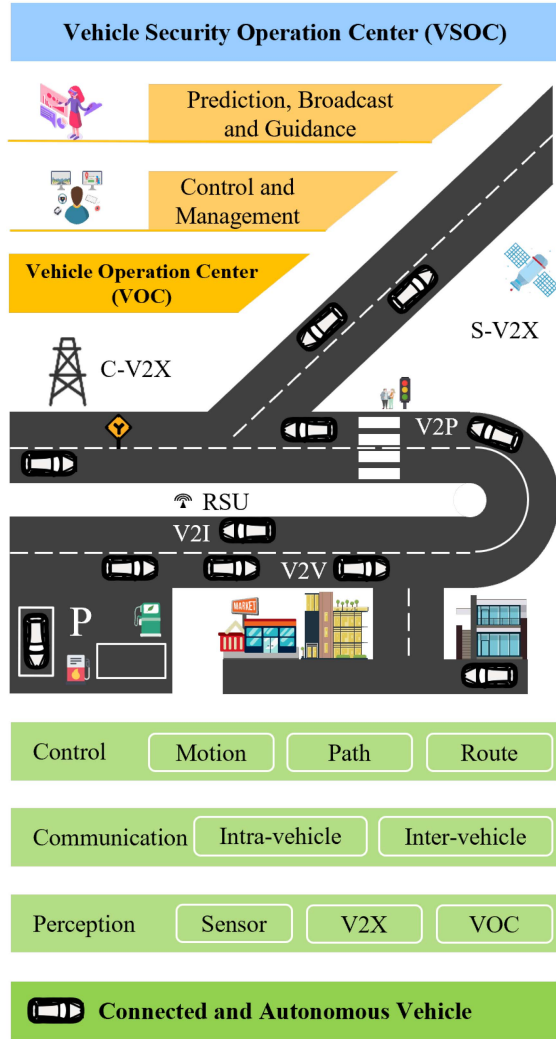


Fig. 1. Connected and autonomous vehicle operation scenario.

3) *Control*: Controlling CAVs to fulfill transportation needs involves both transporting people and goods from one place to another. CAVs require path planning, path tracking [23], and motion control [24] capabilities to navigate from the origin to the target point.

B. Vehicle Operation Center

The VOC controls and manages vehicle resources and driving information prediction, broadcast, and guidance services in CAVs. User trip and cargo transportation needs and road traffic resources constrain the vehicle resources of CAVs. VOC optimizes vehicle production, and scheduling plans to create a safe, green, and economical transportation environment. The vehicle's autonomous driving capability is supported by the in-vehicle system and the VOC's remote coordination and large-scale computing. In addition, VOC provides all kinds of life services for users and makes CAV a super assistant for users. At present, progress in the in-vehicle system by automobile and technology companies has spurred the development of the VOC.

In the past in-vehicle systems relied on GPS and communication systems to provide users with manual telephone-style services (e.g., OnStar, G-Book). Recently, vehicles with autonomous driving capabilities (e.g., Autopilot, XPILOT) have uploaded valid data to the VOC to optimize the model during autonomous driving. The optimized autopilot model will be distributed to vehicles subscribed to the service, making the CAVs driving on the road even better.

III. POTENTIAL SECURITY THREATS TO CONNECTED AND AUTONOMOUS VEHICLES

The communication needs of connected autonomous vehicles will expose them to cybersecurity risks. Identifying and classifying cybersecurity risks is a prerequisite for prevention, and existing classification and modelling approaches have advantages and disadvantages. Sun et al. classified them as intra-vehicle network attacks, V2X network attacks, and other attacks and threats [14]. Ju et al. analyzed intra-vehicle network attacks, sensor attacks, and inter-vehicle attacks from the perspective of CAVs control [12]. The classification of CAVs security threats into internal and external entities facilitates understanding the location of network attack occurrences but may overlook the specific characteristics of cybersecurity threats and attacks. Microsoft classifies cyber threats and attacks from their common elements into six categories: spoofing, tampering, repudiation, information disclosure, denial-of-service, and elevation privilege (STRIDE) [25]. STRIDE is a bottom-up categorization that ignores the relationship between different categories, where an attack belongs to two different types. In contrast, we categorize the cybersecurity threats of CAVs top-down based on the cybersecurity principles of CIA [26] as BC, TD, FD, IT, and PE. And Table II provides a summary of the security threats related to CAVs.

A. Block Communication (BC)

The simplest cyberattack technique is to block communication of CAVs, such as denial-of-service attack (DoS), flooding attack, jamming attack, and black hole attack.

1) *Denial-of-Service Attack*: DoS makes the communication system of CAVs completely unavailable [27]. In connected vehicles, various types of communication occur within the vehicle, between V2V, and between V2I, which are highly vulnerable to DoS attacks [28], [65]. In a DoS attack, the network resources cannot process the requests further, thus making the network resources unavailable to the users. Distributed DoS attacks (DDoS) are more advanced forms of DoS attacks, where the attack can be launched from several different systems to a specific target and disrupt its functionality [66].

2) *Flooding Attack*: A flooding attack is a form of DoS that maliciously uses communication resources to block normal communication activities during communication establishment and usage [30]. In CAVs, the flooding attack attacks the extra-vehicle network (e.g., AODV, DSRC) and the intra-vehicle network (e.g., CAN). The data used in the attack can be divided into two kinds of communication protocol specified data and useless

TABLE II
SUMMARY OF SECURITY THREATS FOR CAVs

Type		Citation	Method	Target Network	Attack Effects
BC	DoS Attack	[27], [28]	Blocking communication using communication protocols or physical vulnerabilities	V2I, V2V, CAN	Making the network resources unavailable to the users
	Flooding Attack	[29]–[31]	Takes up communication resources	AODV, DSRC, CAN	Blocking normal communication
	Jamming Attack	[32]–[36]	Physical attacks on communication lines using radio frequency, etc.	V2I, V2V	Obstruction of wireless communication
	Black Hole Attack	[37]	Dropping all messages passing through a malicious node	V2I, V2V	The target cannot receive the message
TD	—	[38], [39]	DoS with incomplete blocking and post-blocking delayed release message	V2I, V2V, CAN	Messages arrive on target with a delay
FD	FDI Attack	[40]	Sending false data to the target network	V2I, V2V, CAN	Disrupting the performance of CAVs, e.g. increasing the risk of collisions
	Spoofing Attack	[7]	Tempting the GPS receiver to migrate from a legitimate signal to a spoofed signal	GPS	Vehicle attends the wrong position
	Sybil Attack	[41]–[43]	Multiple identity forgery in communication networks	V2V	Malicious nodes predominate in decision making
	Replay Attack	[44]–[47]	Sending the recorded messages to the communication network	CAN	Blocking communication or misleading behavior of CAVs
IT	MITM Attack	[48]–[52]	Middle node eavesdropping on communication messages	V2I, V2V, RFID	Messages sent to the receiver are dropped or modified
	Location Trailing	[53], [54]	Data mining through location information of CAVs	V2V, GPS	Stealing users' privacy and sensitive information
PE	Malware	[55]–[59]	Malware gains partial authority over the device	V2I, V2V	Privacy breaches or attack devices on the network
	ECU Software Flashing Attack	[60], [61]	ECU code rewriting via intra-vehicle network or OTA technology	V2I, CAN	Stealing vehicle information or gaining authority over a vehicle
	Phish Attack	[62]–[64]	Deceiving users into browsing malicious information	V2I, V2V, CAN	Stealing private information or delivering malware

data [29]. The former will consume the communication computing power, and the latter will monopolize the communication channel resources. In addition, using identification to attack or impersonate communication nodes can make the flooding attack more difficult to be detected [31].

3) *Jamming Attack*: The jamming attack disrupts the wireless/radio communication channel by sending jamming signals to block the network communication of V2X [32], [33]. Jamming attacks, also considered a type of DoS attack, exploit physical communication flaws, while other DoS attacks commonly exploit vulnerabilities in the network. Jammer attacks that occur continuously in an area with a single jamming signal are easily detected [34]. Sending jamming signals intermittently is called a periodic jamming attack [35], while sending jamming signals on a channel after identifying the target is known as a reactive jamming attack [36].

4) *Black Hole Attack*: The black hole attack (also known as packet drop attack) by dropping packets in the network that reaches the destination causes the network services within the CAVs to be blocked. The attacker drops some of the packets selectively or randomly is known as the gray hole attack [37].

B. Time Delay (TD)

Time delay attacks are intended to delay the delivery of vehicle information. The impact of this attack is especially

significant when targeting information that requires high real-time performance. Time-delay attacks on location and speed information in the inter-vehicle network will result in traffic deceleration, congestion, and reduced throughput [38], [67]. Delayed transmission of sensing and control information in the intra-vehicle network will result in loss of vehicle control and collision [39].

C. False Data (FD)

The false data is propagation in the CAVs, violating the integrity security requirements. We take false data injection, spoofing attack, sybil attack and replay attack as typical examples to concretely introduce the attack path, method, and effectiveness of false data. All four attacks can be through the intra-vehicle and inter-vehicle networks. At the same time, false data injection focuses on stealthiness. In comparison, spoofing attacks and sybil attacks focus on impersonating the identity of sensors or vehicles.

1) *False Data Injection*: False data injection (FDI) degrades the operational performance of CAVs by sending false sensor data to remote or local controllers through the communication network. An attacker can send false data directly to the CAN via OBD-II [40]. However, the general false data fails to fool the controller and is readily detected by the detector [68]. Hence, attackers are persistently exploring covert FDI attack

methods [69]. These methods seek to bolster CAV security by generating adversarial samples that outwit defensive strategies.

2) *Spoofing Attack*: Spoofing attacks that impersonate or replace actual sensing data in existing communication protocols. The most representative case is the spoofing attack around GPS, which misleads the receiver to acquire erroneous position data by constructing a stronger GPS signal. In addition, the attacker continuously corrects the malicious position for the CAV to arrive at the wrong location [7]. Getting inside the CAN through OBD-II to impersonate a CAN ID such as rpm and gear to issue the abnormal sensing signal is also an important research direction in spoofing attacks [70].

3) *Sybil Attack*: The act of launching an attack by using a pseudonymous vehicle identity is called sybil attack [41], [42]. The messages broadcasted by the pseudonymous vehicles will cause virtual chaos in the traffic area or use the pseudonymous vehicles to influence traffic decisions. Sybil attack does not have a fixed limit on the number of pseudonymous vehicles, which means that the attacker releases multiple pseudonymous vehicles to carry out the attack and even achieve the effect of DoS attack [43]. In addition, various sybil attack variants exist, like the node simulation attack, where simulated nodes send fake messages or malware, protecting the attacker's identity.

4) *Replay Attack*: The replay attack is the attacker records the control and sensor data and then releases it back to the communication network after a while to mislead the CAVs [44], [45]. In the intra-vehicle network, CAN lacks encryption and authentication mechanisms, which makes CAVs vulnerable to replay attacks. Attackers can eavesdrop and perform replay attacks through OBD ports or wireless interfaces connected to CAN [46], [47]. Attackers can also launch replay attacks when they know the key or password of the V2X [71].

D. Information Theft (IT)

Information theft is a type of cyberattack that obtains vehicle or user information from CAVs by exploiting vulnerabilities and data mining methods.

1) *Man-in-the-Middle Attack*: Man-in-the-middle attack (MITM) in which the attacker eavesdrops on the communication and then tries to delete or modify the content of the information forwarded to the receiver [48]. The premise of MITM is that an attacker places a malicious node between two communicating parties and gains trust in the communication between them. Several studies have been conducted to discuss the vulnerable points in CAVs that are susceptible to attack [49], [50]. The transparent authentication method of Passive Entry Passive Start (PEPS) technology for keyless entry is easy for attackers to build relays to execute MITM attacks [51]. Moreover, in the model of using blockchain to store vehicle data, an attacker replaces the blockchain account address of the actual user and can launch a MITM attack [52].

2) *Location Trailing Attack*: The attackers of this attack obtain the privacy and sensitive information of the vehicle owners by mining the location data of the CAV [14]. The location information transmission of CAVs is an essential guarantee for efficient and secure transportation, and the vehicle pseudonym

approach is usually used to protect user privacy [53]. However, attackers can also compromise the vehicle pseudonym approach by exploiting vehicle botnets [54].

E. Privilege Escalation (PE)

We present the following cyberattack methods from the perspective of software, hardware, and social engineering to acquire privileges.

1) *Malware*: Malware (malicious software) gets access to devices and obtains personal and driving data in CAVs devices, causing privacy leaks or being used to attack devices on the network. Malware has many different types [56], such as viruses, worms, Trojan horses, etc. Malware originated from the study of computer networks [55] and entered CAVs with the development of cyberspace. In addition to infrastructure such as RSU and electric vehicle supply equipment (EVSE) in V2I spreading malware to vehicles [56], [57], devices such as cell phones in Vehicle-to-Device (V2D) are also a vital attack surface for vehicle malware [58]. In CAVs' cybersecurity research, blocking the rapid spread of malware is crucial [59].

2) *ECU Software Flashing Attack*: The ECU controls the driving status and other functions of the CAV, and there are different ECUs, such as Engine Management System (EMS), Transmission Control Unit (TCU), and Body Control Module (BCM), depending on their functions. The attacker can rewrite the ECU software to gain control of the CAV using the intra-vehicle network or Over-the-Air technology (OTA) [60]. This type of attack is called an ECU software flashing attack [61].

3) *Phishing Attack*: The phishing attack is a social engineering attack that deceives users or staff of CAVs to leak sensitive information or install malware [62]. Since people are essential participants in CAVs, they have access to networks or devices. The attacker can use e-mail, SMS and other communication tools to release aliased URLs or fraudulent messages [63]. The infotainment system of CAVs [72] integrates e-mail and other communication functions, allowing the attacker to gain permission and information about the vehicle system, making vehicle security highly vulnerable [64].

IV. CYBERSECURITY COUNTERMEASURES FOR CONNECTED AND AUTONOMOUS VEHICLES

We collect the countermeasures of cybersecurity considering the vehicle lifecycle. As shown in Fig. 3, it is necessary to make efforts from standards, communication authentication, and decision verification scheme from the development stage of the vehicle; In the operation stage, CAVs operation needs resilient strategy and attack detection to resist the impact of cyberattacks; In the accidental end-of-life stage, following the cyberattack, timely digital forensics is initiated to obtain the basis for VSOC optimization. And Table III summarizes these security countermeasures.

A. Standardization Efforts (SE)

Cybersecurity needs a unified standard to regulate manufacturers and service providers for product standardization, to

TABLE III
SUMMARY OF CYBERSECURITY COUNTERMEASURES FOR CAVS

Type		Citation	Method	Corresponding Threat
SE	International Standards	[73]–[75]	ISO introduces the vehicle functional safety standard ISO 26262 and the cybersecurity standard ISO 21434, while IEEE presents IEEE 1906.2, which specifies security messages on MAWM	BC, TD, FD, IT, PE
	American Standards	[76]	SAE has released the vehicle cybersecurity standard SAE J3061 and the V2V security protocol standard SAE 2945/1	BC, TD, FD, IT
	European Standards	[77], [78]	ETSI-TC-ITS-WG5 is responsible for data protection and security of ITS	BC, TD, FD, IT
	Chinese Standards	[79]	SAC/TC 260 released GB/T 38628 which presents security standards for automotive electronic systems from a lifecycle perspective	BC, TD, FD, PE
A&V	Communication Authentication	[80]–[85]	Communication authorization provides CAVs with a trusted communication channel	IT
	Decision Verification	[86]–[92]	Decision validation ensures the availability and security of CAVs functions	BC, TD, FD
RS	Resilient Estimation	[93]–[98]	Resilient estimation include attack detection-based state estimation methods and resilient state estimators	BC, TD, FD
	Secure Estimation	[99]–[101]	Secure estimation exploits the redundancy of sensor measurements to accurately estimate the states	BC, TD, FD
	Attack Tolerant Control	[102]–[105]	Controller parameters and event-triggered mechanism (ETM) are designed to mitigate the impact of DoS attacks	BC, TD, FD
	Attack Compensation	[106], [107]	Upon estimation of the attacks, the controller design compensates for the associated influence and mitigates the resultant impact of attacks	BC, TD, FD
AD	Observer-based AD	[108]–[112]	Anomaly detectors based on Luenberger-like observers and unknown input observers are designed for attack detection	BC, TD, FD
	Filer-based AD	[113]–[120]	Attack detection algorithms based on Kalman filters, UFIR filters, and set-membership filters	BC, TD, FD
	Active AD	[121]–[123]	Actively injecting supplementary signals into the system inputs for attack detection	BC, TD, FD
DF	—	[124]–[132]	DF conducts investigations of cyberattacks on vehicles in four phases: readiness planning, data acquisition and preservation, analysis, and reporting	BC, TD, FD, PE

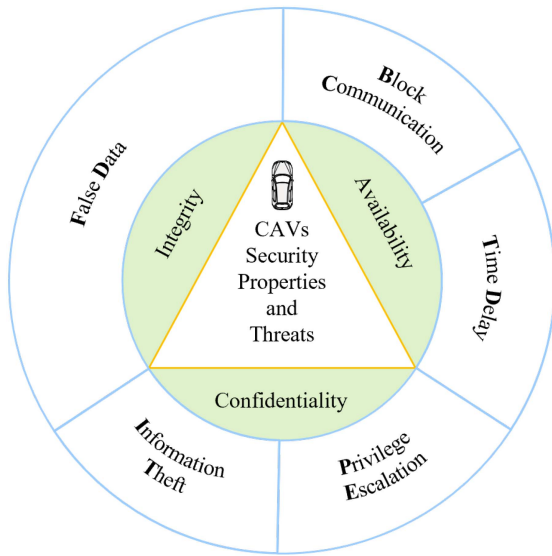


Fig. 2. Visualization of considered security properties and threats for CAVs.

provide the most basic cybersecurity services for CAVs users, and to increase the threshold and difficulty of attacks. In addition to international standards, different regions have different standards, and we introduce the existing standards from global to regional.

1) *International Standards*: International standards organizations include the International Organization for Standardization (ISO) and the Institute of Electrical and Electronics-

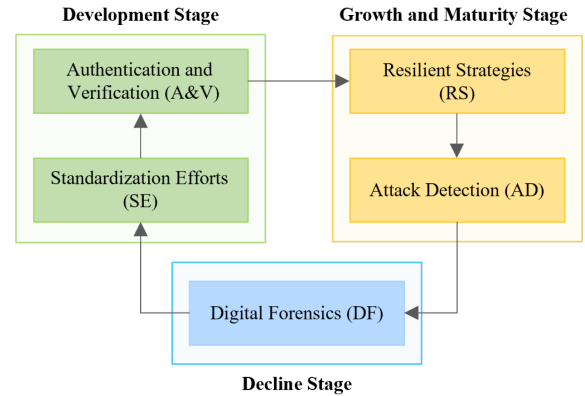


Fig. 3. Taxonomy of cybersecurity countermeasures in the CAVs.

Engineers (IEEE), with standards such as ISO 26262, ISO 21434, and IEEE 1609.2. Based on International Electrotechnical Commission (IEC) 61508, ISO 26262 is an international standard for road vehicle functional safety [73]. First issued in 2011 and updated in 2018, it is the initial mass-produced electrical and electronic system product safety standard. It assigns Automotive Safety Integrity Levels (ASIL) to hazards, reflecting risk reduction levels needed to avert specific dangers. ASIL ranges from D (highest hazard) to A (lowest). The ASIL hazard class is determined by cross-cutting the severity, exposure, and controllability classifications. Furthermore, the ISO standard for road vehicle cybersecurity is ISO 21434 [74], which is more

TABLE IV
ETSI ITS SECURITY STANDARDS [77]

Standard reference	Title
TR 102 893	Threat, Vulnerability and Risk Analysis (TVRA) technical report
TS 102 731	Security services and architecture
TS 103 097	Security header and certificate formats
TS 102 940	ITS communications security architecture and security management
TS 102 941	Trust and privacy management
TS 102 942	Access control
TS 102 943	Confidentiality services

focused on external cyberattacks to protect user property and privacy than ISO 26262. ISO 21434 presents cybersecurity risk management requirements for road vehicles and their components, interfaces, etc., defining the vehicle lifecycle, including vehicle engineering, production, operation, maintenance and decommissioning related to each phase. IEEE 1609 is a higher standard based on IEEE 802.11p, while IEEE 1609.2 is the IEEE 1609 family of protocols describing vehicle security and privacy [75]. IEEE 1609.2 defines secure data structures, secure message formats and the processing of these secure messages within the WAVE (Wireless Access in Vehicular Environments) platform.

2) *American Standards*: American standards organizations are the SAE (Society of Automotive Engineers), SAE Vehicle Cybersecurity Systems Engineering Committee, and DSRC Technical Committee, working groups on cybersecurity and privacy protection for CAVs. SAE J3061 is the predecessor of ISO 21434, and the SAE J3061 can be applied to automotive functions such as ASIL, which means that it can serve as a system cybersecurity guide that ISO 26262 lacks. The standards for vehicle-to-vehicle safety communication systems are SAE J2945/1 [76], which satisfies the needs of on-board systems to ensure that the exchange of Basic Safety Message (BSM) in V2V safety communication guarantees interoperability and data integrity to support the performance of the envisaged safety applications.

3) *European Standards*: European standards organizations include ETSI (European Telecommunications Standards Institute), and the ETSI-TC-ITS-WG5 working group is the main working group on cyber security. The group's security process consists of three steps: (i) identify intelligent transport systems (ITS) security threats, (ii) construct security requirements and devise countermeasures, and (iii) specify an architecture and a standardized set to enable secure V2V and V2I [78]. They have also developed several standard documents in Table IV.

4) *Chinese Standards*: Chinese standards organization is SAC (Standardization Administration of China) and has a national information security standardization technical committee (SAC/TC 260) responsible for cybersecurity standardization. SAC/TC 260 has developed the standard GB/T 38628-2020 [79], which refers to the lifecycle for information security management of ISO 26262, but does not mention ASIL-related content.

B. Authentication and Verification (A&V)

Communication authentication and decision verification ensure the communication security and operational safety of CAVs.

1) *Communication Authentication*: Communication authentication is a security mechanism that validates vehicle communication data identifiers to distinguish legitimate data from malicious data [133]. Traditional authentication methods encompass password, key-based authentication, and biometrics, widely employed across the Internet. However, traditional methods encounter limitations in vehicle-centric CAVs [13]. In particular, authentication technology in this context necessitates low energy consumption, stable operation, and high verification accuracy. The lack of authentication mechanisms in the CAN, a critical part of intra-vehicle networks, poses a significant vulnerability for cyberattacks. Some researchers have embedded cryptographic payloads into the data or identifier field [80], [134], and new security mechanisms (e.g., secure session keys [81]) have been introduced. The inter-vehicle network has higher computing resources and can authenticate messages using public-key infrastructure (PKI) [82], attribute-based encryption (ABE) [83], message authentication code (MAC) [84], and blockchain [85] technologies.

2) *Decision Verification*: The communication system sends the correct perception data to the control unit for the CAVs to make decisions. To guarantee the performance of autonomous driving, the decision-making behavior requires extensive validation. Scenario-based testing has the advantages of flexible configuration, safety, and low cost, making it an integral part of decision validation [86], [87], [88], [89]. Then, for the system software and hardware of CAVs, rarely faults are injected to test the robustness of decision-making and error handling ability under faults [90], [91]. Moreover, a mathematical model of the vehicle decision state and traffic environment is constructed to logically verify the correctness of the decision [92].

C. Resilient Strategies (RS)

The resilient strategies are generally considered from a control perspective, which focuses on maintaining the system performance to some degree under attacks.

1) *Resilient Estimation*: Resilient estimation approaches address the relative accurate estimation issue of CAV systems under sensor measurement attacks. These approaches mainly comprise two methods: attack detection-based method [93], [94], [95] and resilient state estimators [96], [97], [98]. The attack detection-based methods disregard the attacked sensor measurements and depend exclusively on the predictions from the system model for state estimation. In contrast, resilient state estimators still incorporate the attacked measurements but assign relatively lower weights to them. The aforementioned resilient estimation approaches exhibit satisfactory performance under sporadic measurement attacks. However, prolonged and persistent attacks may result in a decline in estimation accuracy.

2) *Secure Estimation*: Secure estimation addresses the state estimation recovery in the presence of sensor measurement

attacks [99], [100], [101]. These techniques exploit the redundancy of sensor measurements to accurately estimate the states of CAVs, even when certain sensors' measurements are corrupted by attack signals. Specifically, the recovery of state estimation is feasible when there are sufficient redundant sensors. It has been demonstrated that the maximum number of attacked sensors should not exceed half of the total number of sensors to ensure reliable state estimation. However, it should be noted that the high degree of sensor redundancy required by secure estimation techniques results in a substantial cost, which limits their practical applicability.

3) *Attack Tolerant Control*: Attack-tolerant controllers are developed by analyzing the stability of CAV systems under attack, with a focus on CAV platoons facing DoS attacks. In the analysis, the maximum allowable attack frequency and attack duration time are generated as functions of the controller parameters [102], [103], [104]. Thus, the parameters can be subsequently adjusted to enhance the resilience of the closed-loop system against attacks. In addition, an event-triggered mechanism (ETM) is developed based on the controllers to mitigate the impact of DoS attacks, which increases communication transmission attempts during attacks [105]. However, it is essential to note that attack-tolerant control strategies are primarily effective against short-term and relatively low-frequency attacks.

4) *Attack Compensation*: The attack compensation resilience methodology is predicated on the underlying principle of estimating attack signals [106], [107]. Upon successful estimation of the attack signal values, the controller design compensates for the associated influence and mitigates the resultant impact of attacks. The attack compensation technique is straightforward and straightforward to implement. Nevertheless, the efficacy of such approaches is profoundly contingent on the accuracy of the attack signal estimation outcomes.

D. Attack Detection (AD)

The attack detection methods mentioned here pertain to anomaly detection from control perspectives, where anomalies resulting from attacks on sensors, intra-vehicle networks and inter-vehicle networks. Three main types of such methods are presented below.

1) *Observer-Based Attack Detection*: Observer-based attack detection method applies to deterministic vehicle dynamics models, where the modelling and measurement uncertainties are assumed to be bounded. The corresponding attack detection is usually designed based on canonical observers. Several observers types are employed in CAV attack detection, encompassing Luenberger-like observers [108], unknown input observers [106], and sliding mode observers [110], [111], [112]. In such approaches, the attacks are usually assumed to be imposed on sensor measurements [108], [112], or the inter-vehicle communications of CAVs [109], [110], [111]. Given the uncertainties of vehicle system models and measurements are bounded, if the residuals corresponding to the observers exceed some properly selected thresholds, the attacks can be detected. designed based on canonical observers for vehicle dynamic systems. Other than the attack detection capability, some of the

designed observers, such as unknown input observers and sliding mode observers, can simultaneously estimate the attack signal.

2) *Filter-Based Attack Detection*: Different from the observer-based attack detection methods, filter-based attack detection methods are suitable for vehicle dynamics models with random modelling uncertainties and measurement noises. These approaches are built based on canonical filters such as Kalman filters/extended Kalman filters [113], [114], [115], [116], [117], improved unbiased finite-impulse response (UFIR) filters [118] and set-membership filters [119], [120]. With the residuals generated from the filters, statistical features of them are extracted to detect anomalies caused by attacks. Multiple approaches process the residuals for attack detection, which are termed detectors. The canonical detectors include χ^2 detector [113], [115], cumulative sum (CUSUM) detector [114], and multivariate exponentially weighted moving average (MEWMA) [113] detector, all of which rely on Kalman filter/extended Kalman filter. The improved detectors include a modified generalized likelihood ratio (GLR) detector [117] and detectors built upon machine learning methods such as support vector machine (SVM) [116] and convolutional neural network (CNN) [115]. By combining the filters and the detectors, attack signals that affect the outputs of CAV systems and inter-vehicle networks are effectively detected.

3) *Active Attack Detection*: Active attack detection techniques involve the injection of supplementary signals into the inputs of Cyber-Physical Systems (CPS). This technique enables the design of attack detection algorithms based on the dissimilarity in residuals between cases with and without attacks, with the aid of observers or filters. A significant advantage of this approach is its ability to uncover stealthy attacks, including the notorious stealthy replay attacks, that are intentionally designed to evade observer/filter-based detection methods. The supplementary signals can take diverse forms, such as Gaussian noise [113], [121], [122] and sinusoidal signals [123], necessitating the development of distinct detection algorithms. For example, statistical features can be used as detectors for CAV systems injected with Gaussian noises [122], while disturbance observers can be employed for those with injected sinusoidal signals [123].

E. Digital Forensics (DF)

Implementing autonomous driving and safe operation for CAVs relies on the support of ECUs, RSUs, and electronic devices of cloud-based systems. These electronic devices are crucial resources for investigating cyberattacks, malfunctions, and accidents. Digital forensics uses ICT to collect, preserve and analyze data from electronic devices to investigate cybercrimes and accidents.

1) *Forensic Guidelines*: As early as 2004, the National Institute of Standards and Technology (NIST) wrote guidance standard SP800-72 for Personal Digital Assistant (PDA) forensics. Subsequently, guidance documents were issued for computer and mobile device forensics. The international standard ISO27037, published in 2012, guides on identifying, collecting, acquiring, and preserving digital evidence. After 2012,

digital evidence was established as independent evidence in China's amended civil, criminal, and administrative litigation laws. Standards Committee TC179/SC7 has developed GB/T 29360-2012, GB/T 29361-2012, and GB/T 29362-2012, which regulate digital forensics from terms definition, operation process, and report format. Moreover, the Ministry of Justice of the People's Republic of China issued a technical specification (SF/T 0077-2020) for electronic data inspection of vehicles to guide vehicle forensic work.

2) *Forensic Process*: Digital forensics typically has four phases: readiness planning, data acquisition and preservation, analysis, and reporting [130].

- *Readiness Planning*: This phase is a proactive process to manage the threat of accidents and cyberattacks before they occur [124]. The goal is to maximize the potential of the digital forensics process with minimal investigative costs. To accomplish this, investigation stakeholders need to be informed of available data resources, collection toolsets, and data selection interfaces. Combining this information, security experts tailor forensic plans for different threat profiles [125].
- *Data Acquisition and Preservation*: Data acquisition is the process of data collection by investigating stakeholders using planned tools [126]. Generally, CAV's physical device preservation and acquisition of all data within the device are considered best practices. In addition to incorporating privacy considerations into collection decisions, investigation stakeholders must weigh the desired objectives against the results of different acquisition methods [127]. Furthermore, CAVs can store data outside the vehicle, e.g., RSU, remote servers. It makes cloud storage of vehicle log files feasible for investigators to access and obtain digital evidence remotely [128]. The tamper-evident feature of blockchain technology provides a technical guarantee to protect the integrity and verifiability of evidence during data acquisition [129].
- *Analysis*: The analysis phase aims to reconstruct the event with a logical and comprehensive evidence trail. CAVs have many data resources but originate from different components with diverse data records. This means that the primary task in this phase is to meet the requirements of repeatability, completeness and consistency of the data [130]. Based on this, comprehensive and reasonable hypotheses are formulated through rules, causality and artificial intelligence methods [131].
- *Reporting*: After the previous investigation, investigation stakeholders eventually completed a written report to communicate any examination results effectively [132]. Often, investigators will write reports per the general requirements in the specifications and guidance for digital forensics. However, there are different reports for different investigative needs, such as technical, investigative, and evaluative reports.

V. VSOC: VEHICLE SECURITY OPERATIONS CENTER

With the technological development of CAVs, vehicle security operations centers (VSOC) will evolve from a supplemental

application of VOC to an independent operational system. In this section, we first present the system components management, orchestration, detection, and response (MODR) of VSOC. Then, we introduce ParaDefender, the basic architecture of VSOC.

A. MODR: System Components of VSOC

VSOC is a vehicle security operations center (SOC) for vehicle cybersecurity. Existing SOC use cybersecurity technology architectures such as endpoint detection and response (EDR), managed detection and response (MDR), extended detection and response (XDR), and security orchestration automation and response (SOAR). After a comprehensive analysis of the above architecture, we clarify the functions and processes of each component in MODR.

1) *Management*: Management is the stage of MODR preparation and post-finishing. The preparation stage commences with the collection of endpoint and communication information for vehicles, RSUs, etc., followed by compliance checks for security standards, authentication and verification of CAVs system information, and the establishment of forensic readiness. Eventually, the application interface is provided for other components. In the finishing stage, the reports of each component are aggregated, statistical analysis of CAVs system alarm and countermeasure information is performed, and the management report of MODR is formed. This component has device management and compliance monitoring similar to EDR and MDR, but the preparation in conjunction with Section IV cybersecurity measures is unique.

2) *Orchestration*: This stage formally describes responding to cybersecurity threats and transforming the complex security operations process into effective workflows. The orchestration has to build a smooth interface between people, people and tools, and tools and tools to form the necessary steps to accomplish the objective of security operations. On the other hand, it needs to optimize the configuration of workflows in different domains that SOAR ignores to achieve specific objectives, such as the budget constraint objective of minimizing operational costs and maximizing application performance.

3) *Detection*: This component is the sense stage of MODR and provides trigger conditions for the orchestrated workflow. From the vehicle control perspective, we can detect anomalous data from observer-based, filter-based, and active attack detection methods, as described in Section IV-D.

4) *Response*: The response is the execution phase of MODR, and its purpose is to minimize the damage. Resilient strategies are essential measures to protect vehicle control from cyber attacks. At the same time, data acquisition and preservation of digital forensics is a critical support to maintain the security of CAVs systems.

B. ParaDefender: Basic Architecture of VSOC

ParaDefender is a cybersecurity defense mechanism we proposed [135], and VSOC is its application in CAV cybersecurity, as shown in Fig. 4.

1) *Three Worlds Three Experts*: Karl Popper illustrates the philosophy that reality consists of physical, mental and artificial worlds. The three ITs of industrial technology, information

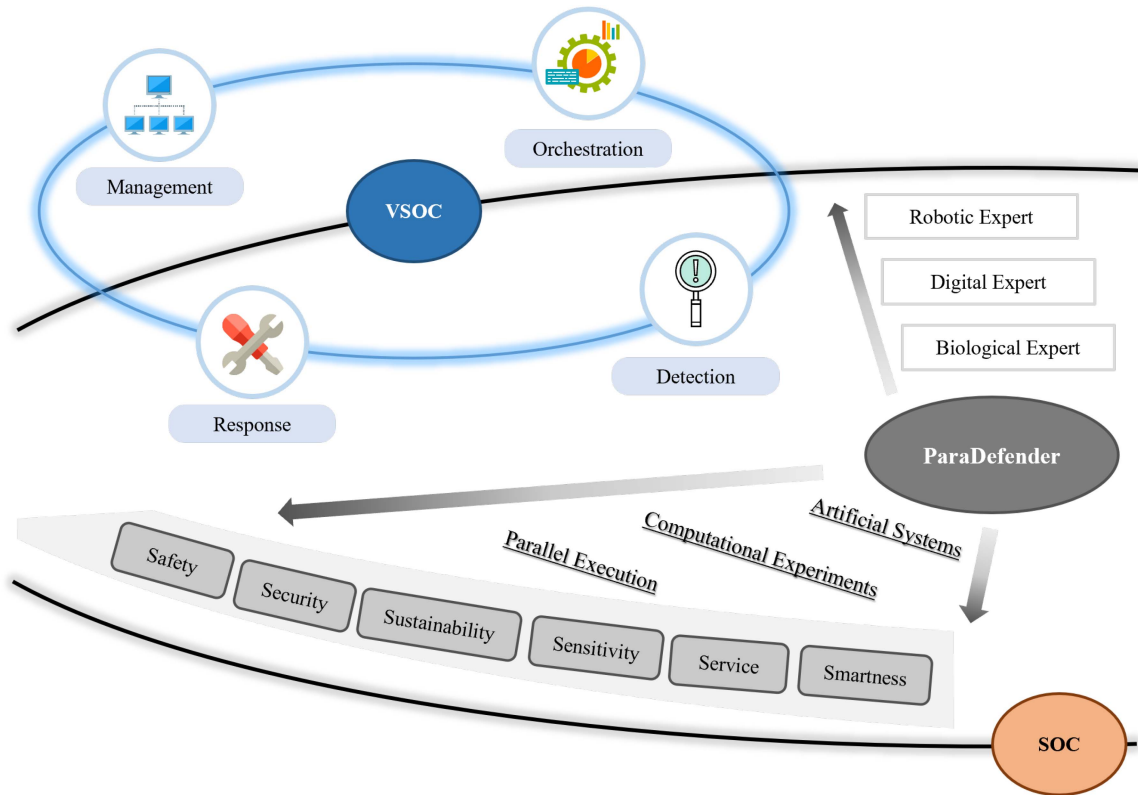


Fig. 4. Visualization of considered security properties and threats for CAVs.

technology and intelligent technology correspond to the development tools of the three worlds. The three ITs develop the whole world of CAVs simultaneously [136]. We believe biological, digital and robotic experts will operate the CAV cybersecurity environment.

2) *Parallel Theory and Methodology*: The artificial system uses scenario engineering to map out multiple possibilities of the actual system, transforming from the traditional model solving to a new perspective of scenario experiment, which gives a foothold to the complex problems of CPSS; Furthermore, the computational experiment explores feasible solutions for network regulation based on the artificial system [137]; Finally, the parallel execution evaluates and verifies the viable solutions, and deploys the optimal solutions to the actual scenario [138]. A large closed loop is formed from actual scenarios to ACP (Artificial system, Computational experiment, Parallel execution) [139] and returns to actual scenarios.

3) *Performance Objectives*: The application of ParaDefender will enable CAV to enter a new realm: physical world security, cyberspace security, overall sustainable development, sensitivity to safeguard privacy and personalized individual development, comprehensive services, and smartness for an intelligence society [140].

C. VSOC Intrinsic Architecture

Fig. 5 illustrates the components of the VSOC intrinsic architecture, comprising artificial VSOC, computational

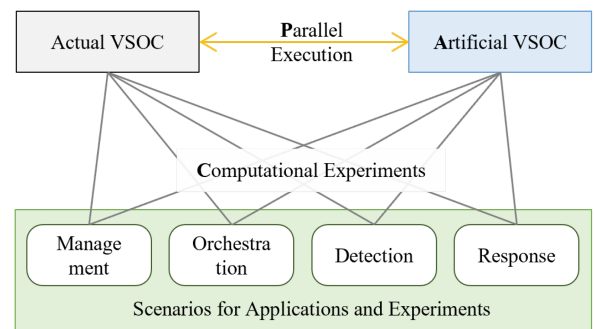


Fig. 5. VSOC intrinsic architecture is designed based on the ACP.

experiments, and parallel execution. The artificial VSOC is a mapping of the actual VSOC and includes all necessary scenario components to construct experimental scenarios based on specific requirements. Computational experiments coordinate experimental scenario scripts and test algorithm optimization schemes to generate feasible solutions for both actual and demanded scenarios. The parallel execution facilitates communication between the actual and artificial VSOCs, deploying the feasible solution and providing effective feedback to the artificial VSOC, forming a closed-loop optimization system.

VI. CONCLUSION

CAVs have become the trend of AVs, and their communication capabilities bring many benefits but also make them vulnerable

to various types of cyberattacks. In this work, we have analyzed various types of network attacks from the principle of network security and multiple countermeasures from the perspective of the security operations lifecycle in detail. However, the current countermeasures are all countermeasure strategies for specific cybersecurity issues, and such strategies can hardly address the increasingly complex cyber environment. Furthermore, CAVs cyberspace is constructed through different protocols and communication objects such as intra-vehicle networks, inter-vehicle networks, and vehicle-road communication networks. Therefore, this work presents a conceptual framework, VSOC, to systematically construe the cybersecurity of CAVs. The framework is guided by the idea of parallel security and consists of management, detection, orchestration, and response as the basic structure of VSOC. We hope that this article will attract attention and discussion in the field of cybersecurity and automotive.

REFERENCES

- [1] L. Chen et al., "Milestones in autonomous driving and intelligent vehicles: Survey of surveys," *IEEE Trans. Intell. Veh.*, vol. 8, no. 2, pp. 1046–1056, Feb. 2023.
- [2] S.-A. Kaye, K. Somoray, D. Rodwell, and I. Lewis, "Users' acceptance of private automated vehicles: A systematic review and meta-analysis," *J. Saf. Res.*, vol. 79, pp. 352–367, Dec. 2021.
- [3] P. Kopelias, E. Demiridi, K. Vogiatzis, A. Skabardonis, and V. Zafiropoulou, "Connected & autonomous vehicles environmental impacts a review," *Sci. Total Environ.*, vol. 712, Apr. 2020, Art. no. 135237.
- [4] Y. Chen, H. Zhang, and F.-Y. Wang, "Society-centered and DAO-Powered sustainability in transportation 5.0: An intelligent vehicles perspective," *IEEE Trans. Intell. Veh.*, vol. 8, no. 4, pp. 2635–2638, Apr. 2023.
- [5] Q. Luo, Y. Cao, J. Liu, and A. Benslimane, "Localization and navigation in autonomous driving: Threats and countermeasures," *IEEE Wireless Commun.*, vol. 26, no. 4, pp. 38–45, Aug. 2019.
- [6] K. Maeng, W. Kim, and Y. Cho, "Consumers' attitudes toward information security threats against connected and autonomous vehicles," *Telematics Inform.*, vol. 63, Oct. 2021, Art. no. 101646.
- [7] S. Liu et al., "All your GPS are belong to us: Towards stealthy manipulation of road navigation systems," in *Proc. 27th USENIX Secur. Symp.*, 2018, pp. 1527–1544.
- [8] M. Hasan, S. Mohan, T. Shimizu, and H. Lu, "Securing vehicle-to-everything (V2X) communication platforms," *IEEE Trans. Intell. Veh.*, vol. 5, no. 4, pp. 693–713, Dec. 2020.
- [9] A. Masood, D. S. Lakew, and S. Cho, "Security and privacy challenges in connected vehicular cloud computing," *IEEE Commun. Surv. Tut.*, vol. 22, no. 4, pp. 2725–2764, Fourthquarter 2020.
- [10] P. Wang, X. Wu, and X. He, "Modeling and analyzing cyberattack effects on connected automated vehicular platoons," *Transp. Res. Part C: Emerg. Technol.*, vol. 115, Jun. 2020, Art. no. 102625.
- [11] R. Gupta, A. Kumari, and S. Tanwar, "A taxonomy of blockchain envisioned edge-as-a-connected autonomous vehicles," *Trans. Emerg. Telecommun. Technol.*, vol. 32, no. 6, 2021, Art. no. e4009.
- [12] Z. Ju, H. Zhang, X. Li, X. Chen, J. Han, and M. Yang, "A survey on attack detection and resilience for connected and automated vehicles: From vehicle dynamics and control perspective," *IEEE Trans. Intell. Veh.*, vol. 7, no. 4, pp. 815–837, Dec. 2022.
- [13] A. Talpur and M. Gurusamy, "Machine learning for security in vehicular networks: A comprehensive survey," *IEEE Commun. Surveys Tut.*, vol. 24, no. 1, pp. 346–379, Firstquarter 2022.
- [14] X. Sun, F. R. Yu, and P. Zhang, "A survey on cyber-security of connected and autonomous vehicles (CAVs)," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 7, pp. 6240–6259, Jul. 2022.
- [15] F. Sommer, J. Dürrewang, and R. Kriesten, "Survey and classification of automotive security attacks," *Information*, vol. 10, no. 4, Apr. 2019, Art. no. 148.
- [16] K. Strandberg, N. Nowdehi, and T. Olovsson, "A systematic literature review on automotive digital forensics: Challenges, technical solutions and data collection," *IEEE Trans. Intell. Veh.*, vol. 8, no. 2, pp. 1350–1367, Feb. 2023.
- [17] J. Yang et al., "Eco-driving of general mixed platoons with CAVs and HDVs," *IEEE Trans. Intell. Veh.*, vol. 8, no. 2, pp. 1190–1203, Feb. 2023.
- [18] A. Zhou, Y. Liu, E. Tenenboim, S. Agrawal, and S. Peeta, "Car-following behavior of human-driven vehicles in mixed-flow traffic: A driving simulator study," *IEEE Trans. Intell. Veh.*, vol. 8, no. 4, pp. 2661–2673, Apr. 2023.
- [19] Y. Qian, X. Wang, Z. Chen, C. Wang, and M. Yang, "Hy-Seg: A hybrid method for ground segmentation using point clouds," *IEEE Trans. Intell. Veh.*, vol. 8, no. 2, pp. 1597–1606, Feb. 2023.
- [20] Y. Li, L. Yang, S. Han, X. Wang, and F.-Y. Wang, "When LPWAN meets ITS: Evaluation of low power wide area networks for V2X communications," in *Proc. IEEE 21st Int. Conf. Intell. Transp. Syst.*, 2018, pp. 473–478.
- [21] Y. Li, S. Han, L. Yang, F.-Y. Wang, and H. Zhang, "LoRa on the move: Performance evaluation of LoRa in V2X communications," in *Proc. IEEE Intell. Veh. Symp.*, 2018, pp. 1107–1111.
- [22] S. Kim and A. S. Ibrahim, "Byzantine-fault-Tolerant consensus via reinforcement learning for permissioned blockchain-empowered V2X network," *IEEE Trans. Intell. Veh.*, vol. 8, no. 1, pp. 172–183, Jan. 2023.
- [23] M. Michalopoulos, P. Kolios, C. Panayiotou, and G. Ellinas, "Optimizing vehicles' tracking accuracies while considering information aging," *IEEE Trans. Intell. Veh.*, vol. 8, no. 3, pp. 2538–2550, Mar. 2023.
- [24] S. Teng et al., "Motion planning for autonomous driving: The state of the art and future perspectives," *IEEE Trans. Intell. Veh.*, vol. 8, no. 6, pp. 3692–3711, Jun. 2023.
- [25] K. Bernsmed, D. S. Cruzes, M. G. Jaatun, and M. Iovan, "Adopting threat modelling in agile software development projects," *J. Syst. Softw.*, vol. 183, Jan. 2022, Art. no. 111090.
- [26] M. K. Khan and A. Quadri, "Augmenting cybersecurity in autonomous vehicles: Innovative recommendations for aspiring entrepreneurs," *IEEE Consum. Electron. Mag.*, vol. 10, no. 3, pp. 111–116, May 2021.
- [27] N. Lyamin, A. Vinel, M. Jonsson, and J. Loo, "Real-time detection of denial-of-service attacks in IEEE 802.11p vehicular networks," *IEEE Commun. Lett.*, vol. 18, no. 1, pp. 110–113, Jan. 2014.
- [28] G. D'Angelo, A. Castiglione, and F. Palmieri, "A cluster-based multidimensional approach for detecting attacks on connected vehicles," *IEEE Internet Things J.*, vol. 8, no. 16, pp. 12518–12527, Aug. 2021.
- [29] N. T. Luong, A. Q. Nguyen, and D. Hoang, "FAPDRP: A flooding attacks prevention and detection routing protocol in vehicular ad hoc network using behavior history and nonlinear median filter transformation," *Wireless Netw.*, vol. 28, pp. 1–28, Oct. 2022.
- [30] H. Karthikeyan and G. Usha, "Real-time DDoS flooding attack detection in intelligent transportation systems," *Comput. Elect. Eng.*, vol. 101, Jul. 2022, Art. no. 107995.
- [31] S. B. Park, H. J. Jo, and D. H. Lee, "Flooding attack mitigator for in-vehicle CAN using fault confinement in CAN protocol," *Comput. Secur.*, vol. 126, Mar. 2023, Art. no. 103091.
- [32] A. Alipour-Fanid, M. Dabaghchian, and K. Zeng, "Impact of jamming attacks on vehicular cooperative adaptive cruise control systems," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 12679–12693, Nov. 2020.
- [33] H. Elghamrawy and A. Noureldin, "Narrowband jamming mitigation based on multi-resolution analysis for land vehicles," *IEEE Trans. Intell. Veh.*, vol. 8, no. 4, pp. 3083–3095, Apr. 2023.
- [34] D. Karagiannis and A. Argyriou, "Jamming attack detection in a pair of RF communicating vehicles using unsupervised machine learning," *Veh. Commun.*, vol. 13, pp. 56–63, Jul. 2018.
- [35] Z. Li, B. Hu, and Z. Yang, "Co-design of distributed event-triggered controller for string stability of vehicle platooning under periodic jamming attacks," *IEEE Trans. Veh. Technol.*, vol. 70, no. 12, pp. 13115–13128, Dec. 2021.
- [36] C. Bajracharya, "Performance evaluation for secure communications in mobile internet of vehicles with joint reactive jamming and eavesdropping attacks," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 11, pp. 22563–22570, Nov. 2022.
- [37] K. M. A. Alheiti, A. Gruebler, and K. McDonald-Maier, "Using discriminant analysis to detect intrusions in external communication for self-driving vehicles," *Digit. Commun. Netw.*, vol. 3, no. 3, pp. 180–187, Aug. 2017.
- [38] Z. Sun, R. Liu, H. Hu, D. Liu, and Z. Yan, "Cyberattacks on connected automated vehicles: A traffic impact analysis," *IET Intell. Transport Syst.*, vol. 17, no. 2, pp. 295–311, 2023.
- [39] X. Li, X. Xu, L. Chen, H. Zhang, Z. Ju, and W. Guo, "Longitudinal speed tracking control for an electric connected vehicle with actuator saturation subject to a replay attack," *Nonlinear Dyn.*, vol. 111, no. 2, pp. 1369–1383, Jan. 2023.

- [40] K. Koscher et al., "Experimental security analysis of a modern automobile," in *Proc. IEEE Symp. Secur. Privacy*, 2010, pp. 447–462.
- [41] J. R. Douceur, "The sybil attack," in *Peer-to-Peer Systems*, P. Druschel, F. Kaashoek, and A. Rowstron, Eds. Berlin, Germany: Springer, 2002, pp. 251–260.
- [42] R. Landa, D. Griffin, R. G. Clegg, E. Mykoniati, and M. Rio, "A sybilproof indirect reciprocity mechanism for peer-to-peer networks," in *Proc. IEEE Conf. Comput. Commun.*, 2009, pp. 343–351.
- [43] M. Baza et al., "Privacy-preserving blockchain-based energy trading schemes for electric vehicles," *IEEE Trans. Veh. Technol.*, vol. 70, no. 9, pp. 9369–9384, Sep. 2021.
- [44] M. Porter, P. Hespanhol, A. Aswani, M. Johnson-Roberson, and R. Vasudevan, "Detecting generalized replay attacks via time-varying dynamic watermarking," *IEEE Trans. Autom. Control*, vol. 66, no. 8, pp. 3502–3517, Aug. 2021.
- [45] X. Xu, X. Li, P. Dong, Y. Liu, and H. Zhang, "Robust reset speed synchronization control for an integrated motor-transmission powertrain system of a connected vehicle under a replay attack," *IEEE Trans. Veh. Technol.*, vol. 70, no. 6, pp. 5524–5536, Jun. 2021.
- [46] R. Islam, R. U. D. Refat, S. M. Yerram, and H. Malik, "Graph-based intrusion detection system for controller area networks," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 3, pp. 1727–1736, Mar. 2022.
- [47] J. Ashraf, A. D. Bakhshi, N. Moustafa, H. Khurshid, A. Javed, and A. Beheshti, "Novel deep learning-enabled LSTM autoencoder architecture for discovering anomalous events from intelligent transportation systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 7, pp. 4507–4518, Jul. 2021.
- [48] V. Kumar, M. Ahmad, D. Mishra, S. Kumari, and M. K. Khan, "RSEAP: RFID based secure and efficient authentication protocol for vehicular cloud computing," *Veh. Commun.*, vol. 22, Apr. 2020, Art. no. 100213.
- [49] A. Irshad, M. Usman, S. A. Chaudhry, H. Naqvi, and M. Shafiq, "A provably secure and efficient authenticated key agreement scheme for energy internet-based vehicle-to-grid technology framework," *IEEE Trans. Ind. Appl.*, vol. 56, no. 4, pp. 4425–4435, Jul./Aug. 2020.
- [50] F. Ahmad, F. Kurugollu, A. Adnane, R. Hussain, and F. Hussain, "MARINE: Man-in-the-middle attack resistant trust model in connected vehicles," *IEEE Internet Things J.*, vol. 7, no. 4, pp. 3310–3322, Apr. 2020.
- [51] P. Razmjouei, A. Kavousi-Fard, M. Dabbaghjamesh, T. Jin, and W. Su, "Ultra-lightweight mutual authentication in the vehicle based on smart contract blockchain: Case of MITM attack," *IEEE Sensors J.*, vol. 21, no. 14, pp. 15839–15848, Jul. 2021.
- [52] H. Shen, J. Zhou, Z. Cao, X. Dong, and K.-K. R. Choo, "Blockchain-based lightweight certificate authority for efficient privacy-preserving location-based service in vehicular social networks," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 6610–6622, Jul. 2020.
- [53] Z. Li, Q. Pei, I. Markwood, Y. Liu, M. Pan, and H. Li, "Location privacy violation via GPS-Agnostic smart phone car tracking," *IEEE Trans. Veh. Technol.*, vol. 67, no. 6, pp. 5042–5053, Jun. 2018.
- [54] M. T. Garip, P. Reiher, and M. Gerla, "BOTVEILLANCE: A vehicular botnet surveillance attack against pseudonymous systems in VANETs," in *Proc. IEEE 11th IFIP Wireless Mobile Netw. Conf.*, 2018, pp. 1–8.
- [55] R. E. Schantz, "BBN's network computing software infrastructure and distributed applications (1970–1990)," *IEEE Ann. Hist. Comput.*, vol. 28, no. 1, pp. 72–88, Jan.–Mar. 2006.
- [56] D. T. Le, K. Q. Dang, Q. L. T. Nguyen, S. Alhelaly, and A. Muthanna, "A behavior-based malware spreading model for vehicle-to-vehicle communications in VANET networks," *Electronics*, vol. 10, no. 19, Jan. 2021, Art. no. 2403.
- [57] S. Mousavian, M. Erol-Kantarci, L. Wu, and T. Ortmeyer, "A risk-based optimization model for electric vehicle infrastructure response to cyber attacks," *IEEE Trans. Smart Grid*, vol. 9, no. 6, pp. 6160–6169, Nov. 2018.
- [58] S. Park and J.-Y. Choi, "Malware detection in self-driving vehicles using machine learning algorithms," *J. Adv. Transp.*, vol. 2020, Jan. 2020, Art. no. e3035741.
- [59] A. Boukerche and Q. Zhang, "Countermeasures against worm spreading: A new challenge for vehicular networks," *ACM Comput. Surv.*, vol. 52, no. 2, May 2019, Art. no. 34.
- [60] L. Yu, J. Deng, R. R. Brooks, and S. B. Yun, "Automobile ECU design to avoid data tampering," in *Proc. 10th Annu. Cyber Inf. Secur. Res. Conf.*, 2015, pp. 1–4.
- [61] G. Bendiab, A. Hameurlaine, G. Germanos, N. Kolokotronis, and S. Shiales, "Autonomous vehicles security: Challenges and solutions using blockchain and artificial intelligence," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 4, pp. 3614–3637, Apr. 2023.
- [62] S. Parkinson, P. Ward, K. Wilson, and J. Miller, "Cyber threats facing autonomous and connected vehicles: Future challenges," *IEEE Trans. Intell. Transp. Syst.*, vol. 18, no. 11, pp. 2898–2915, Nov. 2017.
- [63] E. U. Soykan, M. Bagriyanik, and G. Soykan, "Disrupting the power grid via EV charging: The impact of the SMS phishing attacks," *Sustain. Energy, Grids Netw.*, vol. 26, Jun. 2021, Art. no. 100477.
- [64] G. Costantino, A. La Marra, F. Martinelli, and I. Matteucci, "CANDY: A social engineering attack to leak information from infotainment system," in *Proc. IEEE 87th Veh. Technol. Conf.*, 2018, pp. 1–5.
- [65] F. Sakiz and S. Sen, "A survey of attacks and detection mechanisms on intelligent transportation systems: VANETs and IoV," *Ad Hoc Netw.*, vol. 61, pp. 33–50, Jun. 2017.
- [66] H. H. R. Sherazi, R. Iqbal, F. Ahmad, Z. A. Khan, and M. H. Chaudary, "DDoS attack detection: A key enabler for sustainable communication in internet of vehicles," *Sustain. Comput. Inform. Syst.*, vol. 23, pp. 13–20, Sep. 2019.
- [67] F. Ahmad, A. Adnane, V. N. L. Franqueira, F. Kurugollu, and L. Liu, "Man-in-The-Middle attacks in vehicular ad-hoc networks: Evaluating the impact of attackers' strategies," *Sensors*, vol. 18, no. 11, Nov. 2018, Art. no. 4040.
- [68] J. Zhou, W. Yang, H. Zhang, W. X. Zheng, Y. Xu, and Y. Tang, "Security analysis and defense strategy of distributed filtering under false data injection attacks," *Automatica*, vol. 138, Apr. 2022, Art. no. 110151.
- [69] G.-Y. Yang and X.-J. Li, "Complete stealthiness false data injection attacks against dynamic state estimation in cyber-physical systems," *Inf. Sci.*, vol. 586, pp. 408–423, Mar. 2022.
- [70] E. Seo, H. M. Song, and H. K. Kim, "GIDS: GAN based intrusion detection system for in-vehicle network," in *Proc. IEEE 16th Annu. Conf. Privacy Secur. Trust*, 2018, pp. 1–6.
- [71] D. He, S. Zeadally, B. Xu, and X. Huang, "An efficient identity-based conditional privacy-preserving authentication scheme for vehicular ad hoc networks," *IEEE Trans. Inf. Forensics Secur.*, vol. 10, no. 12, pp. 2681–2691, Dec. 2015.
- [72] D. Kenjić, D. Živkov, and M. Antić, "Automated data transfer from ADAS to android-based IVI domain over SOME/IP," *IEEE Trans. Intell. Veh.*, vol. 8, no. 4, pp. 3166–3177, Apr. 2023.
- [73] T. Stolte et al., "Taxonomy to unify fault tolerance regimes for automotive systems: Defining fail-operational, fail-degraded, and fail-safe," *IEEE Trans. Intell. Veh.*, vol. 7, no. 2, pp. 251–262, Jun. 2022.
- [74] D. Zelle, C. Plappert, R. Rieke, D. Scheuermann, and C. Krauß, "Threat-Surf: A method for automated threat surface assessment in automotive cybersecurity engineering," *Microprocessors Microsystems*, vol. 90, Apr. 2022, Art. no. 104461.
- [75] M. A. R. Baee, L. Simpson, E. Foo, and J. Pieprzyk, "Broadcast authentication in latency-critical applications: On the efficiency of IEEE 1609.2," *IEEE Trans. Veh. Technol.*, vol. 68, no. 12, pp. 11577–11587, Dec. 2019.
- [76] J. B. Kenney, "Dedicated short-range communications (DSRC) standards in the United States," *Proc. IEEE*, vol. 99, no. 7, pp. 1162–1182, Jul. 2011.
- [77] B. Linc and P. Cincilla, "Cooperative ITS security framework: Standards and implementations progress in Europe," in *Proc. IEEE 17th Int. Symp. A World Wireless Mobile Multimedia Netw.*, 2016, pp. 1–6.
- [78] B. Fernandes, J. Rufino, M. Alam, and J. Ferreira, "Implementation and analysis of IEEE and ETSI security standards for vehicular communications," *Mobile Netw. Appl.*, vol. 23, no. 3, pp. 469–478, Jun. 2018.
- [79] *Information Security Technology—Guide for Automotive Electronics Systems*, SAC Standard GB/T 38628-2020, Apr. 2020.
- [80] B. Groza, L. Popa, and P.-S. Murvai, "Highly efficient authentication for CAN by identifier reallocation with ordered CMACs," *IEEE Trans. Veh. Technol.*, vol. 69, no. 6, pp. 6129–6140, Jun. 2020.
- [81] B. Palaniswamy, S. Camtepe, E. Foo, and J. Pieprzyk, "An efficient authentication scheme for intra-vehicular controller area network," *IEEE Trans. Inf. Forensics Secur.*, vol. 15, pp. 3107–3122, 2020.
- [82] M. A. Simplicio, E. L. Cominetti, H. K. Patil, J. E. Ricardini, L. T. D. Ferraz, and M. V. M. Silva, "Privacy-preserving certificate linkage/revocation in VANETs without linkage authorities," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 6, pp. 3326–3336, Jun. 2021.
- [83] J. Cui, X. Chen, J. Zhang, Q. Zhang, and H. Zhong, "Toward achieving fine-grained access control of data in connected and autonomous vehicles," *IEEE Internet Things J.*, vol. 8, no. 10, pp. 7925–7937, May 2021.
- [84] H. Zhong, W. Cao, Q. Zhang, J. Zhang, and J. Cui, "Toward trusted and secure communication among multiple internal modules in CAV," *IEEE Internet Things J.*, vol. 8, no. 24, pp. 17734–17746, Dec. 2021.
- [85] P. Kumar, R. Kumar, G. P. Gupta, and R. Tripathi, "BDEdge: Blockchain and deep-learning for secure edge-envisioned green CAVs," *IEEE Trans. Green Commun. Netw.*, vol. 6, no. 3, pp. 1330–1339, Sep. 2022.

- [86] X. Li, S. Teng, B. Liu, X. Dai, X. Na, and F.-Y. Wang, "Advanced scenario generation for calibration and verification of autonomous vehicles," *IEEE Trans. Intell. Veh.*, vol. 8, no. 5, pp. 3211–3216, May 2023.
- [87] J. Wang et al., "Parallel vision for long-tail regularization: Initial results from IVFC autonomous driving testing," *IEEE Trans. Intell. Veh.*, vol. 7, no. 2, pp. 286–299, Jun. 2022.
- [88] C. Neurohr, L. Westhofen, T. Henning, T. de Graaff, E. Möhlmann, and E. Böde, "Fundamental considerations around scenario-based testing for automated driving," in *Proc. IEEE Intell. Veh. Symp.*, 2020, pp. 121–127.
- [89] J. Dong et al., "Mixed cloud control testbed: Validating vehicle-road-cloud integration via mixed digital twin," *IEEE Trans. Intell. Veh.*, vol. 8, no. 4, pp. 2723–2736, Apr. 2023.
- [90] Y. Fu, A. Terechko, T. Bijlsma, P. J. L. Cuijpers, J. Redegeld, and A. O. Örs, "A retargetable fault injection framework for safety validation of autonomous vehicles," in *Proc. IEEE Int. Conf. Softw. Architecture Companion*, 2019, pp. 69–76.
- [91] S. Jha et al., "ML-Based fault injection for autonomous vehicles: A case for Bayesian fault injection," in *Proc. IEEE/IFIP 49th Annu. Int. Conf. Dependable Syst. Netw.*, 2019, pp. 112–124.
- [92] T. Zhao, E. Yurtsever, J. A. Paulson, and G. Rizzoni, "Formal certification methods for automated vehicle safety assessment," *IEEE Trans. Intell. Veh.*, vol. 8, no. 1, pp. 232–249, Jan. 2023.
- [93] R. G. Dutta, F. Yu, T. Zhang, Y. Hu, and Y. Jin, "Security for safety: A path toward building trusted autonomous vehicles," in *Proc. IEEE/ACM Int. Conf. Comput.-Aided Des.*, 2018, pp. 1–6.
- [94] R. G. Dutta, "Security of autonomous systems under physical attacks: With application to self-driving cars," Ph.D. dissertation, Univ. Central Florida, Orlando, FL, USA, 2018.
- [95] Q. Liu, Y. Mo, X. Mo, C. Lv, E. Mihankhah, and D. Wang, "Secure pose estimation for autonomous vehicles under cyber attacks," in *Proc. IEEE Intell. Veh. Symp.*, 2019, pp. 1583–1588.
- [96] X. Liu, R. Jiang, H. Wang, and S. S. Ge, "Filter-based secure dynamic pose estimation for autonomous vehicles," *IEEE Sensors J.*, vol. 19, no. 15, pp. 6298–6308, Aug. 2019.
- [97] R. G. Dutta, Y. Hu, F. Yu, T. Zhang, and Y. Jin, "Design and analysis of secure distributed estimator for vehicular platooning in adversarial environment," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 4, pp. 3418–3429, Apr. 2022.
- [98] F. Yu, R. G. Dutta, T. Zhang, Y. Hu, and Y. Jin, "Fast attack-resilient distributed state estimator for cyber-physical systems," *IEEE Trans. Comput.-Aided Des. Integr. Circuits Syst.*, vol. 39, no. 11, pp. 3555–3565, Nov. 2020.
- [99] Y. Shoukry and P. Tabuada, "Event-triggered state observers for sparse sensor noise/attacks," *IEEE Trans. Autom. Control*, vol. 61, no. 8, pp. 2079–2091, Aug. 2016.
- [100] M. Showkatbakhsh, Y. Shoukry, R. H. Chen, S. Diggavi, and P. Tabuada, "An SMT-based approach to secure state estimation under sensor and actuator attacks," in *Proc. IEEE 56th Annu. Conf. Decis. Control*, 2017, pp. 157–162.
- [101] Y. Shoukry, P. Nuzzo, A. Puggelli, A. L. Sangiovanni-Vincentelli, S. A. Seshia, and P. Tabuada, "Secure state estimation for cyber-physical systems under sensor attacks: A satisfiability modulo theory approach," *IEEE Trans. Autom. Control*, vol. 62, no. 10, pp. 4917–4932, Oct. 2017.
- [102] R. Merco, F. Ferrante, and P. Pisu, "DoS-Resilient hybrid controller for string-stable connected vehicles," in *Proc. IEEE Intell. Veh. Symp.*, 2019, pp. 1639–1644.
- [103] R. Merco, F. Ferrante, and P. Pisu, "A hybrid controller for DoS-Resilient string-stable vehicle platoons," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 3, pp. 1697–1707, Mar. 2021.
- [104] D. Zhang, Y. P. Shen, S. Q. Zhou, X. W. Dong, and L. Yu, "Distributed secure platoon control of connected vehicles subject to DoS attack: Theory and application," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 51, no. 11, pp. 7269–7278, Nov. 2021.
- [105] V. S. Dolk, P. Tesi, C. De Persis, and W. P. M. H. Heemels, "Event-triggered control systems under denial-of-service attacks," *IEEE Trans. Control Netw. Syst.*, vol. 4, no. 1, pp. 93–105, Mar. 2017.
- [106] Q. Chen, Y. Zhou, S. Ahn, J. Xia, S. Li, and S. Li, "Robustly string stable longitudinal control for vehicle platoons under communication failures: A generalized extended state observer-based control approach," *IEEE Trans. Intell. Veh.*, vol. 8, no. 1, pp. 159–171, Jan. 2023.
- [107] G. Savaia, Z. Abdollahi Biron, and P. Pisu, "A receding horizon switching control resilient to communication failures for connected vehicles," in *Proc. Dyn. Syst. Control Conf.*, 2017, pp. 1–6.
- [108] X. He, E. Hashemi, and K. H. Johansson, "Distributed control under compromised measurements: Resilient estimation, attack detection, and vehicle platooning," *Automatica*, vol. 134, Dec. 2021, Art. no. 109953.
- [109] E. Khanapuri, T. Chintalapati, R. Sharma, and R. Gerdes, "Learning based longitudinal vehicle platooning threat detection, identification and mitigation," *IEEE Trans. Intell. Veh.*, vol. 8, no. 1, pp. 290–300, Jan. 2023.
- [110] T. Keijzer and R. M. G. Ferrari, "A sliding mode observer approach for attack detection and estimation in autonomous vehicle platoons using event triggered communication," in *Proc. IEEE 58th Conf. Decis. Control*, 2019, pp. 5742–5747.
- [111] X. Zhang, H. Du, J. Wei, Z. Jia, S. Jia, and G. Ma, "High gain observer design for DoS attack detection in CACC platoon," in *Proc. IEEE Int. Seminar Intell. Technol. Appl.*, 2020, pp. 254–259.
- [112] Z. A. Biron and P. Pisu, "Sensor and actuator fault detection in connected vehicles under a packet dropping network," *Int. J. Inf. Commun. Eng.*, vol. 10, no. 6, pp. 1114–1120, 2016.
- [113] M. Porter, A. Joshi, P. Hespanho, A. Aswani, M. Johnson-Roberson, and R. Vasudevan, "Simulation and real-world evaluation of attack detection schemes," in *Proc. IEEE Amer. Control Conf.*, 2019, pp. 551–558.
- [114] R. Quinonez, J. Giraldo, L. Salazar, E. Bauman, A. Cardenas, and Z. Lin, "SAVIOR: Securing autonomous vehicles with robust physical invariants," in *Proc. 29th USENIX Secur. Symp.*, 2020, pp. 895–912.
- [115] Z. He, Y. Chen, H. Zhang, and D. Zhang, "WKN-OC: A new deep learning method for anomaly detection in intelligent vehicles," *IEEE Trans. Intell. Veh.*, vol. 8, no. 3, pp. 2162–2172, Mar. 2023.
- [116] Y. Wang, N. Masoud, and A. Khojandi, "Real-time sensor anomaly detection and recovery in connected automated vehicle sensors," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 3, pp. 1411–1421, Mar. 2021.
- [117] Z. Ju, H. Zhang, and Y. Tan, "Distributed deception attack detection in platoon-based connected vehicle systems," *IEEE Trans. Veh. Technol.*, vol. 69, no. 5, pp. 4609–4620, May 2020.
- [118] Z. Ju, H. Zhang, and Y. Tan, "Deception attack detection and estimation for a local vehicle in vehicle platooning based on a modified UFIR estimator," *IEEE Internet Things J.*, vol. 7, no. 5, pp. 3693–3705, May 2020.
- [119] E. Mousavinejad, F. Yang, Q. L. Han, Q. Qiu, and L. Vlacic, "Cyber attack detection in platoon-based vehicular networked control systems," in *Proc. IEEE 27th Int. Symp. Ind. Electron.*, 2018, pp. 603–608.
- [120] E. Mousavinejad, F. Yang, Q. L. Han, X. Ge, and L. Vlacic, "Distributed cyber attacks detection and recovery mechanism for vehicle platooning," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 9, pp. 3821–3834, Sep. 2020.
- [121] P. Hespanhol, M. Porter, R. Vasudevan, and A. Aswani, "Sensor switching control under attacks detectable by finite sample dynamic watermarking tests," *IEEE Trans. Autom. Control*, vol. 66, no. 10, pp. 4560–4574, Oct. 2021.
- [122] M. Porter, P. Hespanhol, A. Aswani, M. Johnson-Roberson, and R. Vasudevan, "Detecting generalized replay attacks via time-varying dynamic watermarking," *IEEE Trans. Autom. Control*, vol. 66, no. 8, pp. 3502–3517, Aug. 2021.
- [123] G. Na and Y. Eun, "Active probing signal-based attack detection method for autonomous vehicular systems," in *Proc. IEEE 20th Int. Conf. Control Automat. Syst.*, 2020, pp. 53–59.
- [124] V. R. Khande and H. S. Venter, "Novel digital forensic readiness technique in the cloud environment," *Australian J. Forensic Sci.*, vol. 50, no. 5, pp. 552–591, Sep. 2018.
- [125] C. Katsini, G. E. Raptis, C. Alexakos, and D. Serpanos, "FoRePlan: Supporting digital forensics readiness planning for internet of vehicles," in *Proc. 25th Pan-Hellenic Conf. Informat.*, 2022, pp. 369–374.
- [126] S. Grigaliunas, J. Toldinas, A. Venckauskas, N. Morkevicius, and R. Damaševičius, "Digital evidence object model for situation awareness and decision making in digital forensics investigation," *IEEE Intell. Syst.*, vol. 36, no. 5, pp. 39–48, Sep./Oct. 2021.
- [127] G. Horsman, "An 'order of data acquisition' for digital forensic investigations," *J. Forensic Sci.*, vol. 67, no. 3, pp. 1215–1220, 2022.
- [128] M. A. Hoque and R. Hasan, "AVGuard: A forensic investigation framework for autonomous vehicles," in *Proc. IEEE Int. Conf. Commun.*, 2021, pp. 1–6.
- [129] A. O. Philip and R. K. Saravanaguru, "Smart contract based digital evidence management framework over blockchain for vehicle accident investigation in IoV era," *J. King Saud Univ. - Comput. Inf. Sci.*, vol. 34, no. 7, pp. 4031–4046, Jul. 2022.
- [130] K. Gomez, "Analysis of digital forensics capabilities on state-of-the-art vehicles," *Master's thesis*, Tech. Univ. Ingolstadt, Ingolstadt, Germany, Dec. 2019.

- [131] G. De La Torre Parra, L. Selvera, J. Khoury, H. Irizarry, E. Bou-Harb, and P. Rad, "Interpretable federated transformer log learning for cloud threat forensics," in *Proc. Netw. Distrib. Syst. Secur. Symp.*, 2022, pp. 1–16.
- [132] G. Horsman, "Technical reporting in digital forensics," *J. Forensic Sci.*, vol. 67, no. 6, pp. 2458–2468, 2022.
- [133] A. Daeinabi and A. G. Rahbar, "Detection of malicious vehicles (DMV) through monitoring in vehicular ad-hoc networks," *Multimedia Tools Appl.*, vol. 66, no. 2, pp. 325–338, Sep. 2013.
- [134] S. Woo, H. J. Jo, and D. H. Lee, "A practical wireless attack on the connected car and security protocol for in-vehicle CAN," *IEEE Trans. Intell. Transp. Syst.*, vol. 16, no. 2, pp. 993–1006, Apr. 2015.
- [135] J. Han et al., "ParaDefender: A scenario-driven parallel system for defending metaverses," *IEEE Trans. Syst., Man, Cybern. Syst.*, vol. 53, no. 4, pp. 2118–2127, Apr. 2023.
- [136] F.-Y. Wang et al., "Where does AlphaGo go: From church-turing thesis to AlphaGo thesis and beyond," *IEEE/CAA J. Automatica Sinica*, vol. 3, no. 2, pp. 113–120, Apr. 2016.
- [137] F.-Y. Wang, "Parallel control: A method for data-driven and computational control: Parallel control: A method for data-driven and computational control," *Acta Automatica Sinica*, vol. 39, no. 4, pp. 293–302, Mar. 2014.
- [138] J. Wang, X. Wang, Y. Tian, Y. Wang, J. Niu, and O. Kwan, "Parallel training: An ACP-Based training framework for iterative learning in uncertain driving spaces," *IEEE Trans. Intell. Veh.*, vol. 8, no. 4, pp. 2832–2841, Apr. 2023.
- [139] J. Chen, Y. Zhang, S. Teng, Y. Chen, H. Zhang, and F.-Y. Wang, "ACP-Based energy-efficient schemes for sustainable intelligent transportation systems," *IEEE Trans. Intell. Veh.*, vol. 8, no. 5, pp. 3224–3227, May 2023.
- [140] F.-Y. Wang, "The DAO to MetaControl for MetaSystems in metaverses: The system of parallel control systems for knowledge automation and control intelligence in CPSS," *IEEE/CAA J. Automatica Sinica*, vol. 9, no. 11, pp. 1899–1908, Nov. 2022.



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